LECTURE 10

GROUP AND RING THEORY

Theorem 7.7. *Let R be a Noetherian ring having no non-zero nilpotent ideals. Then R has no non-zero nil ideals.*

Proof. (Non-examinable for the oral exam) Suppose on the contrary that $N \neq 0$ is a nil ideal in R. Let
$$\mathcal{F} = \{\mathrm{Ann}_R(n) \mid n \in N\backslash\{0\}\}$$
be a family of left annihilator ideals. Since R is Noetherian, by Theorem 6.13(iii) there exists $m \in N\backslash\{0\}$ such that $\mathrm{Ann}_R(m)$ is a maximal element in $\mathcal{F}$.

Let $x \in R$. Then $mx \in N$, so there exists a smallest $k \in \mathbb{N}$ such that $(mx)^k = 0$. We claim that $mxm = 0$, and assuming the claim, we then get $(RmR)^2 = R(mRRm)R = 0$. Therefore, by hypothesis, we obtain $RmR = 0$. Then $m = 1m1 = 0$, a contradiction. So $N = 0$ in this case. Suppose now that $1 \notin R$. Then we consider the ideal
$$(m) = mR + Rm + RmR + m\mathbb{Z}$$
generated by m; here $RmR = \{\sum_{i=1}^{\ell} r_i m s_i \mid \ell \in \mathbb{N},\, r_i, s_i \in R\}$ (see Exercise Sheet 11). Set $A = mR + RmR$. As $mxm = 0$ for all $x \in R$, it follows that $A^2 = 0$. Hence by hypothesis $A = 0$. Thus $(m) = Rm + m\mathbb{Z}$. Since N is nil, there is a $\ell \in \mathbb{N}$ such that $m^\ell = 0$. Then $(Rm + m\mathbb{Z})^\ell = 0$ and again $Rm + m\mathbb{Z} = (m) = 0$, which yields $m = 0$, a contradiction.

It remains to establish the claim. Indeed, if $k = 1$, then $mx = 0$ gives $mxm = 0$. If $k \geq 2$, observe that
$$\mathrm{Ann}_R(m) \subseteq \mathrm{Ann}_R((mx)^{k-1}).$$
Since $(mx)^{k-1} \neq 0$, we obtain $\mathrm{Ann}_R((mx)^{k-1}) \in \mathcal{F}$. Then by the maximality of $\mathrm{Ann}_R(m)$, we have
$$\mathrm{Ann}_R(m) = \mathrm{Ann}_R((mx)^{k-1}).$$
Now
$$(mx)^k = 0 \quad \Longrightarrow \quad mx \in \mathrm{Ann}_R((mx)^{k-1}) = \mathrm{Ann}_R(m) \quad \Longrightarrow \quad mxm = 0,$$
as claimed. □

Here is an alternative proof to the above, where we actually prove a stronger statement, namely: *Let R be a Noetherian ring having no non-zero nilpotent ideals. Then R has no non-zero nil left ideals.*

Proof. Suppose on the contrary that $A \neq 0$ is a left nil ideal. In particular $A^2 \neq 0$, and therefore, since $A^2 \subseteq AR$, there is an $a \in A$ such that $aR \neq 0$.

Write $U := aR \neq 0$. Observe that $U = \{ax \mid x \in R\}$ is a right nil ideal, since $xa \in A$ implies that $(xa)^{n-1} = 0$ for some $n \in \mathbb{N}$. Therefore $(ax)^n = a(xa)^{n-1}x = 0$.

Now define
$$\mathcal{F} = \{\mathrm{Ann}_R(u) \mid u \in U\backslash\{0\}\}.$$
Since R is Noetherian, there exists $u_0 \in U\backslash\{0\}$ such that $\mathrm{Ann}_R(u_0)$ is a maximal element of $\mathcal{F}$.

Note that $\mathrm{Ann}_R(u_0) \subseteq \mathrm{Ann}_R(u_0x)$ for each $x \in R$. Also, for each $y \in R$, there exists $k \in \mathbb{N}$ such that $(u_0y)^k = 0$ but $(u_0y)^{k-1} \neq 0$. As $(u_0y)^{k-1} = u_0\big(y(u_0y)^{k-2}\big)$, upon setting $x := y(u_0y)^{k-2}$, we see that $u_0x \neq 0$. Maximality then implies that

$$\mathrm{Ann}_R(u_0) = \mathrm{Ann}_R(u_0x) = \mathrm{Ann}_R((u_0y)^{k-1}).$$

Thus $u_oy \in \mathrm{Ann}_R((u_0y)^{k-1}) = \mathrm{Ann}_R(u_0)$, which gives $u_0yu_0 = 0$ for each $y \in R$. Then we claim that Ru_0R is a nilpotent ideal, since

$$(Ru_0R)^2 = R(u_0RRu_0)R = 0 \qquad \Longrightarrow \qquad Ru_0R = 0.$$

We now set

$$B = \{s \in R \mid RsR = 0\}.$$

Note that $B^3 = 0$ and that $0 \neq u_0 \in B$. So B is a non-zero nilpotent ideal. This is the desired contradiction. □

Next we show that a nil ideal in a Noetherian ring is nilpotent.

Theorem 7.8. *Let N be a nil ideal in a Noetherian ring R. Then N is nilpotent.*

Proof. Let T be the sum of nilpotent ideals in R. Then R/T has no non-zero nilpotent ideals, for if A/T is nilpotent, then $(A/T)^m = 0$ implies $(A^m + T)/T = 0$. Hence $A^m \subseteq T$. However, since by Lemma 7.6 the ideal T is nilpotent, there exists $k \in \mathbb{N}$ such that $(A^m)^k = 0$. Hence A itself is nilpotent, so $A \subseteq T$. This implies $A/T = 0$. [An alternative to this part, is to instead let T be a maximal nilpotent (2-sided) ideal. Then R/T has no non-zero nilpotent ideals, for if A/T is nilpotent, then $(A/T)^m = 0$ implies $(A^m + T)/T = 0$. Hence $A^m \subseteq T$. However, since T is nilpotent, we get that there is an $n \in \mathbb{N}$ such that $(A^m)^n \subseteq T^n = 0$. Hence A is nilpotent and since $T \subseteq A$ by assumption, we get $A = T$.]

Consider now the nil ideal $(N + T)/T$ in R/T. By Theorem 7.7, we have $(N + T)/T = 0$. This implies $N \subseteq T$, which is a nilpotent ideal. Hence N is nilpotent. □

Before proving a major result concerning Artinian rings, we need the following. Recall that for a ring R, an *idempotent* is an element $a \in R$ such that $a^2 = a$.

Lemma 7.9. *Let A be a minimal left ideal in a ring R. Then either $A^2 = 0$ or $A = Re$, where e is an idempotent in R.*

Proof. Suppose $A^2 \neq 0$. Then there exists $a \in A$ such that $Aa \neq 0$. However $Aa \subseteq A$, and the minimality of A gives $Aa = A$. Then there exists $e \in A$ such that $ea = a$, and clearly $e \neq 0$ because $a \neq 0$. Moreover $e^2a = ea$, equivalently $(e^2 - e)a = 0$.

As $Re \subseteq A$, it suffices to show that $Re \neq 0$. To this end, we will show that $e^2 - e = 0$, as then $0 \neq e = e^2 \in Re$. Consider $B = \{c \in A \mid ca = 0\}$. Then B is a left ideal of R such that $B \subseteq A$ and $B \neq A$ (since $e \in A \backslash B$). Therefore, by the minimality of A, we must have $B = 0$. Then $(e^2 - e)a = 0$ implies $e^2 = e$, and we are done. □

The following theorem gives useful structural information on Artinian rings with unity and no non-zero nilpotent ideals. Recall that by "unity" we mean non-zero unity.

Theorem 7.10 (Wedderburn–Artin)**.** *Let R be a left (or right) Artinian ring with unity and no non-zero nilpotent ideals. Then R is isomorphic to a finite direct sum of matrix rings over division rings.*

Proof. The proof is broken up into first establishing the following claims:

Claim 1: Each non-zero left ideal in R is of the form Re for some idempotent e.

Claim 2: $R = Re_1 \oplus \cdots \oplus Re_n$ is a sum of minimal left ideals.

Then assuming these two claims, in the family of minimal left ideals $Re_1, \ldots, Re_n$, choose a largest subfamily consisting of all minimal left ideals that are not isomorphic to each other as left R-modules. After renumbering if necessary, let this subfamily be $Re_1, \ldots, Re_k$.

For a fixed $1 \leq j \leq n$, suppose the number of left ideals in the family $(Re_i)_{1\leq i\leq n}$ that are isomorphic to Re_j is n_j. Then

$$R = \underbrace{(Re_1 \oplus \cdots)}_{n_1 \text{ summands}} \oplus \underbrace{(Re_2 \oplus \cdots)}_{n_2 \text{ summands}} \oplus \cdots \oplus \underbrace{(Re_k \oplus \cdots)}_{n_k \text{ summands}},$$

where each set of brackets contains pairwise isomorphic minimal left ideals, and no minimal left ideal in any pair of brackets is isomorphic to a minimal left ideal in another pair. By observing that $\mathrm{Hom}_R(Re_i, Re_j) = 0$ for $i \neq j$ with $1 \leq i, j \leq k$, and recalling Schur's Lemma that $\mathrm{Hom}_R(Re_i, Re_i) = D_i$, a division ring, we obtain, by invoking Theorem 4.30,

$$\begin{aligned}
&\mathrm{Hom}_R(R,R)\\
&\cong \begin{pmatrix} \begin{pmatrix} D_1 & \cdots & D_1 \\ \vdots & & \vdots \\ D_1 & \cdots & D_1 \end{pmatrix}_{n_1\times n_1} & & & \\ & \begin{pmatrix} D_2 & \cdots & D_2 \\ \vdots & & \vdots \\ D_2 & \cdots & D_2 \end{pmatrix}_{n_2\times n_2} & & \\ & & \ddots & \\ & & & \begin{pmatrix} D_k & \cdots & D_k \\ \vdots & & \vdots \\ D_k & \cdots & D_k \end{pmatrix}_{n_k\times n_k} \end{pmatrix}\\
&= \begin{pmatrix} \mathrm{M}_{n_1}(D_1) & & & \\ & \mathrm{M}_{n_2}(D_2) & & \\ & & \ddots & \\ & & & \mathrm{M}_{n_k}(D_k) \end{pmatrix}\\
&\cong \mathrm{M}_{n_1}(D_1) \oplus \mathrm{M}_{n_2}(D_2) \oplus \cdots \oplus \mathrm{M}_{n_k}(D_k).
\end{aligned}$$

However, since

- $\mathrm{Hom}_R(R,R) \cong R^{\mathrm{op}}$ as rings (similar to Example 4.29),
- the opposite ring of a finite direct sum is the direct sum of the opposite rings of the respective summands (the best way to see this is to use the fact that a finite direct sum is isomorphic to a finite direct product),
- the opposite ring of a division ring is a division ring (exercise),
- $(\mathrm{M}_n(D))^{\mathrm{op}} \cong \mathrm{M}_n(D^{\mathrm{op}})$ for D a division ring (see Lecture 0),

the ring R is a finite direct sum of matrix rings over division rings.

So it remains to prove the two claims, which we now do.

Proof of Claim 1: Let A be any non-zero left ideal. By the DCC on left ideals in R, the left ideal A contains a minimal left ideal M. By Lemma 7.9, either $M^2 = 0$ or $M = Re$ for some idempotent e. If $M^2 = 0$, then

$$(MR)^2 = M(RM)R \subseteq MMR = M^2R = 0,$$

i.e. MR is a nilpotent ideal. By hypothesis, we have $MR = 0$, which gives $M = 0$, a contradiction. Hence $M = Re$. Therefore each non-zero left ideal contains a non-zero idempotent.

Then set

$$\mathcal{F} = \{R(1-e') \cap A \mid e' \text{ is a non-zero idempotent in } A\},$$

which is a family of left ideals. Clearly $\mathcal{F}$ is non-empty, since $0 \in \mathcal{F}$. Since R is left Artinian, the set $\mathcal{F}$ has a minimal member, say $R(1-e) \cap A$.

Subclaim: $R(1-e) \cap A = 0$. Otherwise, there exists a non-zero idempotent $f \in R(1-e) \cap A$. As $(1-e)e = 0$, we have $fe = 0$. Set $\widetilde{e} = e + f - ef$. One can verify that $\widetilde{e}\,\widetilde{e} = \widetilde{e}$ and $f\widetilde{e} \neq 0$. Indeed,

$$\begin{aligned}
\widetilde{e}\,\widetilde{e} &= (e+f-ef)(e+f-ef) \\
&= e^2 + ef - e^2f + fe + f^2 - fef - efe - ef^2 + efef \\
&= e + ef - ef + f - ef \quad (\text{using } fe = 0 \text{ and } e^2 = e,\ f^2 = f) \\
&= e + f - ef \\
&= \widetilde{e}
\end{aligned}$$

and so $\widetilde{e}$ is an idempotent. Further

$$f\widetilde{e} = f(e+f-ef) = fe + f^2 - fef = f \neq 0.$$

Next, as

$$1 - \widetilde{e} = (1-e) - f + ef = (1-e) - f(1-e) \in R(1-e),$$

(also $f \in R(1-e)$) we have

$$R(1-\widetilde{e}) \subseteq R(1-e)$$

and so

$$R(1-\widetilde{e}) \cap A \subseteq R(1-e) \cap A.$$

Furthermore we observe that $f \in \big(R(1-e) \cap A\big) \setminus \big(R(1-\widetilde{e}) \cap A\big)$, as if $f \in R(1-\widetilde{e}) \cap A$, then for some $x \in R$, we have $f\widetilde{e} = x(1-\widetilde{e})\widetilde{e} = 0$, a contradiction to $f\widetilde{e} \neq 0$. Therefore, we have

$$R(1-\widetilde{e}) \cap A \subsetneq R(1-e) \cap A,$$

a contradiction to the minimality of $R(1-e) \cap A$ in $\mathcal{F}$. This establishes our subclaim that $R(1-e) \cap A = 0$. (The proof of this subclaim is non-examinable for the oral exam.)

Now, let $a \in A$. Then, recalling that $e \in A$, we have $a(1-e) = a - ae \in R(1-e) \cap A = 0$. Thus $a = ae \in Re$. Then

$$A \subseteq Re \subseteq A$$

proves that $A = Re$, as asserted.

Proof of Claim 2: Let S be the sum of all minimal left ideals in R. We aim to show that $S = R$.

From Claim 1, as S is a non-zero left ideal, we have $S = Re$ for some idempotent e. If $R(1-e) \neq 0$, then there exists a minimal left ideal A contained in $R(1-e)$. Of course then $A \subseteq S = Re$. Thus

$$A \subseteq Re \cap R(1-e).$$

We claim that $Re \cap R(1-e) = 0$. Indeed, if $x \in Re \cap R(1-e)$, then

$$x = ae = b(1-e) \quad \text{for some } a, b \in R.$$

Hence

$$xe = aee = ae = x,$$

and so

$$x = xe = b(1-e)e = b(e-e^2) = 0.$$

Thus

$$A \subseteq Re \cap R(1-e) = 0,$$

a contradiction. Hence $R(1-e) = 0$, and therefore

$$1(1-e) = 1 - e = 0 \quad \Longrightarrow \quad e = 1.$$

Thus

$$S = Re = R1 = R = \sum_{i \in \Lambda} A_i,$$

where $(A_i)_{i\in\Lambda}$ is the family of all minimal left ideals in R. By Theorem 5.9 there exists a subfamily $(A_i)_{i\in\Lambda'}$, of the family of the minimal left ideals such that $R = \oplus_{i\in\Lambda'} A_i$ (which is not necessarily a finite direct sum). Let $1 = e_{i_1} + \cdots + e_{i_n}$, for $0 \neq e_{i_j} \in A_{i_j}$ with $i_j \in \Lambda'$. Then $R = Re_{i_1} \oplus \cdots \oplus Re_{i_n}$, which is a finite direct sum. After reindexing if necessary, we may write $R = Re_1 \oplus \cdots \oplus Re_n$, a direct sum of minimal left ideals.

The proof is now complete.

[An alternative approach instead of (most of) Claim 1 and all of Claim 2 is the following.

First observe that, since R has no non-zero nilpotent ideals (using the first paragraph of Claim 1), by the previous lemma, every minimal left ideal of R is of the form Re for some idempotent e.

Then consider

$$\mathcal{F} = \left\{ \bigcap_{i=1}^{n} R(1-e_i) \mid R = Re_1 \oplus \cdots \oplus Re_n \oplus \bigcap_{i=1}^{n} R(1-e_i) \text{ with all } Re_i \text{ minimal left ideals} \right\}.$$

The set $\mathcal{F}$ is non-empty, since from Exercise 5.2 we know that $R = Re \oplus R(1-e)$ for any idempotent e, and we can choose e such that Re is a minimal left ideal. So $\mathcal{F}$ has a minimal element. Suppose this minimal element $\bigcap_{i=1}^{n} R(1-e_i)$ is non-zero. Consider a minimal left ideal $0 \neq Re_{n+1}$ of $\bigcap_{i=1}^{n} R(1-e_i)$. Then, as $R = Re_{n+1} \oplus R(1-e_{n+1})$,

$$\begin{aligned}
\bigcap_{i=1}^{n} R(1-e_i) &= \bigcap_{i=1}^{n} R(1-e_i) \bigcap \big(Re_{n+1} \oplus R(1-e_{n+1})\big) \\
&= Re_{n+1} \oplus \bigcap_{i=1}^{n+1} R(1-e_i) \qquad \text{by the modular law.}
\end{aligned}$$

As $\bigcap_{i=1}^{n+1} R(1-e_i) \subseteq \bigcap_{i=1}^{n} R(1-e_i)$, by minimality we must have $\bigcap_{i=1}^{n+1} R(1-e_i) = 0$. This now contradicts the fact that the minimal element of $\mathcal{F}$ is non-zero. Therefore the minimal element of $\mathcal{F}$ is 0. Hence $R = Re_1 \oplus Re_2 \oplus \cdots \oplus Re_n$.] $\square$

NOTE: The rest of this chapter is non-examinable for the oral exam.

Note that one can extend the definition of a vector space over a field, to a vector space over a division ring, using the same axioms. Hence, for D a division ring, we have that $\mathrm{M}_n(D)$ is an n^2-dimensional vector space over D, and each left ideal as well as each right ideal of $\mathrm{M}_n(D)$ is a subspace over D. Thus, as in Example 6.14, $\mathrm{M}_n(D)$ is both a Noetherian and an Artinian ring.

As matrix rings over division rings are both right and left Noetherian and Artinian, and a finite direct sum of Noetherian and Artinian rings is again Noetherian and Artinian (Corollary 6.11), we get the following.

Theorem 7.11. *Let R be a left (or right) Artinian ring with unity and no non-zero nilpotent ideals. Then R is also right and left Artinian, and right and left Noetherian.*

Not every ring that is Artinian on one side is Artinian on the other side. The following example gives a ring that is left Artinian but not right Artinian.

Example 7.12. Let $R = \begin{pmatrix} \mathbb{Q} & \mathbb{Q} \\ 0 & 0 \end{pmatrix}$ and consider $A = \begin{pmatrix} 0 & \mathbb{Q} \\ 0 & 0 \end{pmatrix} \subseteq R$. Now A is an ideal of R, and as a left R-module it is simple. Thus A is left Artinian. Also

$$R/A \cong \begin{pmatrix} \mathbb{Q} & 0 \\ 0 & 0 \end{pmatrix}$$

is a field. Therefore R/A as an R/A-module (or as an R-module) is Artinian. Since A is a left Artinian R-module, we obtain that R as a left R-module is Artinian (Theorem 6.10); that is, R is a left Artinian ring, but R is not right Artinian. For, there exists a strictly descending chain

$$\begin{pmatrix} 0 & 2\mathbb{Z} \\ 0 & 0 \end{pmatrix} \supsetneq \begin{pmatrix} 0 & 2^2\mathbb{Z} \\ 0 & 0 \end{pmatrix} \supsetneq \begin{pmatrix} 0 & 2^3\mathbb{Z} \\ 0 & 0 \end{pmatrix} \supsetneq \cdots$$

of right ideals in R.

The following is an important application of the Wedderburn–Artin Theorem. Recall the group algebra from Example 6.15(i).

Theorem 7.13 (Maschke). *If G is a finite group, then*

$$\mathbb{C}[G] \cong \mathrm{M}_{n_1}(\mathbb{C}) \oplus \cdots \oplus \mathrm{M}_{n_k}(\mathbb{C})$$

for some $n_1, \ldots, n_k \in \mathbb{N}$.

Proof. We first prove that $\mathbb{C}[G]$ has no non-zero nilpotent ideals. Let $G = \{g_1 = e, g_2, \ldots, g_n\}$, and $x = \sum \alpha_i g_i \in \mathbb{C}[G]$. Set $x^* = \sum \overline{\alpha}_i g_i^{-1}$, where $\overline{\alpha}_i$ denotes the complex conjugate of α_i. Then

$$xx^* = \sum_{i=1}^{n} |\alpha_i|^2 + \sum_{i=2}^{n} \beta_i g_i$$

for some $\beta_i \in \mathbb{C}$. Hence $xx^* = 0$ implies $\sum_{i=1}^{n} |\alpha_i|^2 = 0$, so each $\alpha_i = 0$ which gives $x = 0$. Now let A be a nilpotent ideal in $\mathbb{C}[G]$, and let $a \in A$. Then $aa^* \in A$, so aa^* is nilpotent, say $(aa^*)^r = 0$ for some $r \in \mathbb{N}$. Without loss of generality, we may assume r is even; since if it is odd, we can just multiply $(aa^*)^r$ by aa^* and still get an element of A, as A is an ideal. Set $b = (aa^*)^{r/2}$. Then $b^2 = 0$ and, as a direct check yields that $(xx^*)^* = xx^*$, we have $b = b^*$. Thus $bb^* = 0$, which gives $(aa^*)^{r/2} = b = 0$. Proceeding like this (i.e. multiplying by aa^* is $r/2$ is odd, and considering the corresponding b for this new power of aa^*), we eventually get $aa^* = 0$. Hence $a = 0$, which proves that $A = (0)$. Hence $\mathbb{C}[G]$ has no non-zero nilpotent ideals.

Further $\mathbb{C}[G]$ is a finite-dimensional algebra with unity over the field $\mathbb{C}$. Therefore by Example 6.15, we have $\mathbb{C}[G]$ is an Artinian ring. Then by the Wedderburn–Artin theorem,

$$\mathbb{C}[G] \cong \mathrm{M}_{n_1}(D^{(1)}) \oplus \cdots \oplus \mathrm{M}_{n_k}(D^{(k)}),$$

where $D^{(i)}$, for $1 \leq i \leq k$, are division rings. Now each $\mathrm{M}_{n_i}(D^{(i)})$ contains a copy K of $\mathbb{C}$ in its centre. In this way each $\mathrm{M}_{n_i}(D^{(i)})$ is a finite-dimensional algebra over K (compare with Example 6.15). Let $[D^{(i)} : K] = n$, and $a \in D^{(i)}$. Then $1, a, a^2, \ldots, a^n$ are linearly dependent over K. Thus, there exist $\alpha_0, \alpha_1, \ldots, \alpha_n$ (not all zero) in K such that $\alpha_0+\alpha_1 a+\cdots+\alpha_n a^n = 0$. However since K is algebraically closed, the polynomial $\alpha_0 + \alpha_1 x + \cdots + \alpha_n x^n \in K[x]$ has all its roots in K. Hence $a \in K$, which shows that $D^{(i)} = K \cong \mathbb{C}$ and completes the proof. □

The next theorem gives some equivalent statements for left Artinian rings that have no non-zero nilpotent ideals. Note that we say that a ring R is *simple* if the only ideals of R are (0) and R itself.

Theorem 7.14. *Let R be a ring with unity. The following are equivalent:*

(i) *R is left Artinian with no non-zero nilpotent ideals;*
(ii) *R is left Artinian with no non-zero nil ideals;*
(iii) *R is a finite direct sum of minimal left ideals;*
(iv) *each left ideal of R is of the form Re, for e an idempotent;*
(v) *R is a finite direct sum of matrix rings over division rings.*

Proof. (ii) $\Longrightarrow$ (i): Follows from the fact that nil ideals in an Artinian ring are nilpotent (Theorem 7.5).

(i) $\Longrightarrow$ (iii): Follows from the proof given for the Wedderburn–Artin Theorem (Theorem 7.10).

(iii) $\Longrightarrow$ (iv): Follows from Exercise Sheet 6, Question 3.

(iv) $\Longrightarrow$ (v): Let $M = Re$ be a maximal left ideal of R with $e = e^2$. Then $R(1-e)$ is a minimal left ideal because $R/Re \cong R(1-e)$ (using the First Isomorphism Theorem) is a simple R-module. Thus if S is the sum of all minimal left ideals of R then $S \neq (0)$. We claim $S = R$. Otherwise S is contained in a maximal left ideal Rf, say, where $f = f^2$. However then $R(1-f)$ is a minimal left ideal which is not contained in S because $S \cap R(1-f) \subseteq Rf \cap R(1-f) = (0)$, a contradiction. Thus $S = R$. Then, as shown in the beginning of Theorem 7.10, the ring R is a finite direct sum of matrix rings over division rings.

(v) $\Longrightarrow$ (i): Since the matrix rings over division rings are right and left Artinian, and the finite direct sum of left and right Artinian rings is again left and right Artinian, we get R is left and right Artinian. Further, the matrix rings over division rings are simple rings with unity (see Exercise Sheet 11) and, hence cannot possess non-zero nilpotent ideals. Now if $R = A_1 \oplus \cdots \oplus A_k$ is a finite direct sum of simple rings each with unity, then it is easy to verify that any non-zero ideal I is of the form $A_{i_1} \oplus \cdots \oplus A_{i_r}$, where $\{i_1, \ldots, i_r\} \subseteq \{1, \ldots, k\}$. Hence I cannot be nilpotent. □

Definition 7.15. A ring satisfying any of these conditions is called a *semisimple Artinian ring.*

Note that Example 7.3 is not a semisimple ring.